

Math708 – Foundations of Computational Mathematics I

Fall 2012

Homework 3

1. For an arbitrary partition $a \leq x_0 < \cdots < x_n \leq b$ of the interval $[a, b]$, let $\mathcal{I}_n(f) = (b-a) \sum_{j=0}^n \sigma_j f(x_j)$ denote a quadrature formula. Show that its degree of exactness is $\leq 2n+1$, i.e., there is a $p \in \mathbb{P}_{2n+2}$ with $\mathcal{I}_n(p) \neq \int_a^b p(x) dx$.
2. Determine the coefficients $a_0, a_1, a_2 \in \mathbb{R}$ after making a comparison with the coefficients to the Taylor series expansion of f . The coefficients should be chosen such that the resulting quadrature formula $Q(f) = a_0 f(a) + a_1 f(\frac{a+b}{2}) + a_2 f(b)$ for the approximation of $\int_a^b f(x) dx$ will have a degree of exactness as high as possible.
3. For a periodic continuous function $f: \mathbb{R} \rightarrow \mathbb{R}$, let Lf denote the interpolating trigonometric polynomial of the form

$$(Lf)(x) = A_0/2 + \sum_{j=1}^N (A_j \cos(jx) + B_j \sin(jx))$$

with support points $x_j = 2\pi j/(2N+1)$, $j = 0, 1, \dots, 2N$, and let $Q(f) = \int_0^{2\pi} (Lf)(x) dx$. Show that $Q(f)$ can be written as $Q(f) = \sum_{j=0}^{2N} a_j f(x_j)$ with positive weights $a_j > 0$ for $j = 0, 1, \dots, 2N$ (independent of f).

4. Determine the coefficients and weights $\{(\lambda_j, \sigma_j)\}_{j=1}^n$ of the Gaussian quadrature formulas $\mathcal{I}_n(f) = \sum_{j=1}^n \sigma_j f(\lambda_j)$ for the approximation of $\int_a^b f(x) \rho(x) dx$ based on the Legendre polynomials and the Chebyshev polynomials for $n = 1, 2, 3$ respectively. Show all details and print them in a table.
5. Prove the Theorem 5.9 in the lecture note.
Theorem 5.9: For $f \in C^{2n}[a, b]$, the error produced using Gaussian quadrature satisfies

$$\begin{aligned} \mathcal{I}(f) - \mathcal{I}_n(f) &= \left[\frac{1}{(2n)!} \int_a^b p_n^2(x) \rho(x) dx \right] f^{(2n)}(\xi) \\ &= \frac{(b-a)^{2n+1}}{(2n)!} \left[\int_0^1 \prod_{k=1}^n (t-t_k)^2 \rho((b-a)t+a) dt \right] f^{(2n)}(\xi) \end{aligned}$$

with $t_k = \frac{\lambda_k - a}{b-a}$ for $k = 1, 2, \dots, n$ and with a suitable intermediate value $\xi \in [a, b]$.

6. (**Programming Exercise**) Numerically calculate the three proper integrals

$$\int_0^{0.5} \frac{1}{16x^2+1} dx, \quad \int_0^2 e^{-x^2} dx, \quad \int_0^{\pi/2} \sqrt{|\cos 2x|} dx$$

by extrapolation of the trapezoidal rule $\mathbf{T}^{h_j}(f)$ using the Romberg step size $h_0 = b - a$, $h_j = h_{j-1}/2$, $j \in \mathbb{N}$ and the Neville's Algorithm (Theorem 4.14 in the lecture notes). More precisely, calculate the quantities $T_{j,\dots,j+m}$ for $j = 0, 1, \dots$, and for $m = 0, \dots, \min\{j, m_*\}$ and stop with $j = j_*$ when

$$“j = 13” \quad \text{or} \quad “m_* + 1 \leq j \leq 12, \quad |T_{j,\dots,j+m_*} - T_{j-1,\dots,j-1+m_*}| \leq \epsilon”$$

is satisfied with $m_* = 4$ and $\epsilon = 10^{-8}$. Provide the values $T_{j,\dots,j+m}$ for $j = 0, 1, \dots, j_*$ and $m = 0, \dots, \min\{j, m_*\}$ in a table for each of the three integrals.